Machine Learning Classification Assignment

Breast Cancer Wisconsin (Diagnostic) Dataset

This report compares five classification models using a fixed stratified 80/20 train-test split. The dataset has 569 instances and 30 numeric input features; the target is malignant (0) or benign (1).

Problem Statement

Implement Logistic Regression, Decision Tree, K-Nearest Neighbors, Gaussian Naive Bayes, and Support Vector Machine on the same dataset and compare Accuracy, AUC, weighted Precision, weighted Recall, weighted F1, and MCC.

Dataset Description

Source: UCI Machine Learning Repository through the copy bundled with scikit-learn. The dataset satisfies the minimum requirement of 500 instances and 12 features.

GitHub Repository Link

[INSERT GITHUB LINK AFTER UPLOAD]

Model Comparison

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9825	0.9954	0.9825	0.9825	0.9825	0.9623
Support Vector Machine	0.9825	0.9967	0.9825	0.9825	0.9825	0.9623
KNN	0.9737	0.9884	0.9747	0.9737	0.9735	0.9442
Naive Bayes	0.9386	0.9878	0.9384	0.9386	0.9384	0.8676
Decision Tree	0.9123	0.9387	0.9137	0.9123	0.9127	0.8139

Performance Observations

Logistic Regression: Achieved accuracy 0.9825, AUC 0.9954, F1 0.9825, and MCC 0.9623.

Support Vector Machine: Achieved accuracy 0.9825, AUC 0.9967, F1 0.9825, and MCC 0.9623.

KNN: Achieved accuracy 0.9737, AUC 0.9884, F1 0.9735, and MCC 0.9442.

Naive Bayes: Achieved accuracy 0.9386, AUC 0.9878, F1 0.9384, and MCC 0.8676.

Decision Tree: Achieved accuracy 0.9123, AUC 0.9387, F1 0.9127, and MCC 0.8139.

Overall Winner

Logistic Regression achieved the highest weighted F1 score on the fixed test set and is selected as the overall winner. The comparison also reports AUC and MCC to avoid relying on accuracy alone.

Streamlit Application

The app accepts a labeled CSV, provides a model-selection dropdown, loads a saved pipeline without retraining, displays all six metrics, and renders a confusion matrix and classification report.

BITS Virtual Lab Screenshot

Insert one authentic screenshot captured while running this project in BITS Virtual Lab. This placeholder cannot serve as proof.